Vendor Performance Analysis Using Python, SQL, and Statistical Techniques

A Comprehensive Data Analysis Report

Made By:

Name: Aman Raj

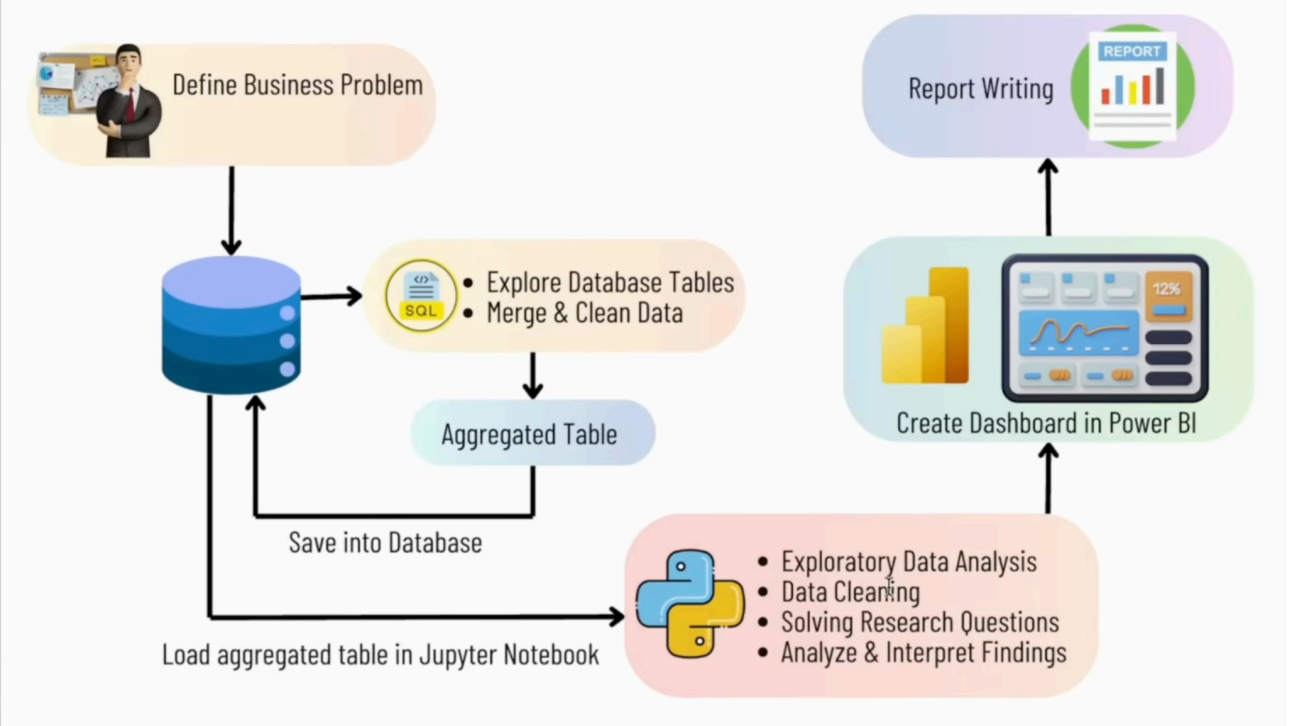
Project Type

Data Analytics Project

Tools & Technologies Used:

Python, SQL (SQLite), Pandas, NumPy, Matplotlib, Seaborn, SciPy, Jupyter Notebook, Power BI (Visualization)

Date: 29 June to 6 July



Executive Summary

This project aims to analyze vendor performance using historical purchase, sales, pricing, and inventory data to support data-driven business decisions. The primary objective is to identify high-performing and underperforming vendors, evaluate inventory efficiency, analyze profitability, and optimize procurement strategies.

The analysis begins by integrating multiple datasets into a consolidated Vendor Sales Summary table using SQL and Python. The data is then cleaned and transformed through preprocessing techniques such as handling missing values, removing inconsistencies, and creating new business metrics. Key performance indicators including Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio are calculated to evaluate vendor performance.

Exploratory Data Analysis (EDA) is performed to identify trends, outliers, correlations, and business patterns. Statistical methods, including confidence interval estimation and hypothesis testing, are applied to validate whether significant profitability differences exist between high-performing and low-performing vendors.

The findings reveal vendor dependency risks, slow-moving inventory, the financial benefits of bulk purchasing, and pricing opportunities for low-performing brands. Based on these insights, actionable recommendations are provided to improve profitability, optimize inventory management, and strengthen procurement strategies.

Business Problem

In the retail and wholesale industry, maintaining the right balance between inventory levels, procurement costs, and sales performance is essential for maximizing profitability. Inefficient inventory management, excessive dependence on a small number of vendors, poor-performing products, and ineffective pricing strategies can lead to increased holding costs, reduced cash flow, and lower overall business performance.

To support data-driven decision-making, this project analyzes vendor and brand performance using sales, purchase, and inventory data. The analysis aims to identify opportunities for improving profitability, optimizing procurement strategies, and enhancing inventory efficiency.

Project Objectives

1. Identify underperforming brands that require promotional campaigns or pricing adjustments.
2. Determine the top vendors contributing to total sales, purchases, and gross profit.

3. Analyze the impact of bulk purchasing on unit costs and potential cost savings.
4. Evaluate inventory turnover to identify slow-moving products and reduce holding costs.
5. Measure vendor dependency and assess the risks associated with over-reliance on a few suppliers.
6. Compare the profitability of high-performing and low-performing vendors using statistical analysis.
7. Provide actionable business recommendations to improve profitability, inventory management, and procurement decisions.

Data Description

The analysis is based on four relational datasets containing information about purchases, sales, pricing, and freight costs. Since the required business metrics are distributed across multiple tables, they were integrated to create a consolidated **Vendor Sales Summary** table for analysis.

1. Purchases Table

The **Purchases** table records all procurement transactions made by vendors. It contains details such as purchase date, vendor information, product (brand), purchase quantity, purchase price, and total purchase amount.

2. Purchase Prices Table

The **Purchase Prices** table stores product pricing information, including the vendor purchase price and the actual product price. Each **Vendor ~ Brand** combination is unique, making this table the primary source for pricing information.

3. Vendor Invoice Table

The **Vendor Invoice** table contains invoice-level information for each purchase order. In addition to purchase quantities and purchase amounts, it includes the **Freight Cost** incurred for each vendor. Each record is uniquely identified by the combination of **Vendor** and **Purchase Order (PO) Number**.

4. Sales Table

The **Sales** table captures actual sales transactions, including the brand sold, quantity sold, selling price, total sales revenue, and applicable excise tax. This dataset is used to evaluate sales performance and profitability.

Data Integration

Since the required analytical information is distributed across multiple tables, a consolidated **Vendor Sales Summary** table was created by joining the four datasets using SQL.

The summary table combines:

- Vendor purchase transactions
- Product sales information
- Freight cost for each vendor
- Actual product prices
- Purchase prices
- Sales revenue
- Purchase quantities
- Sales quantities

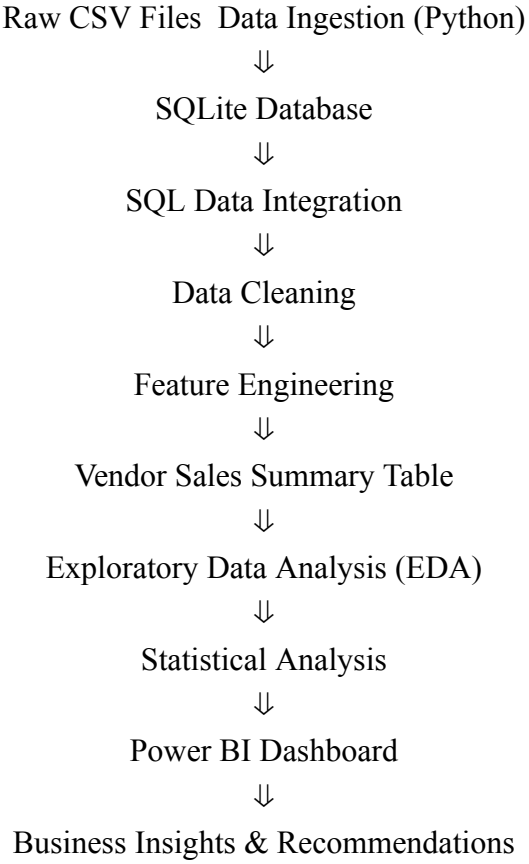
Feature Engineering

To support business analysis, several additional metrics were derived from the consolidated dataset:

Feature	Formula	Purpose
Gross Profit	Total Sales − Total Purchase	Measures profit earned
Profit Margin (%)	(Gross Profit / Total Sales) × 100	Measures profitability
Stock Turnover	Sales Quantity / Purchase Quantity	Evaluates inventory efficiency
Sales_to_Purchase Ratio	Sales Dollars / Purchase Dollars	Compares sales against procurement costs

Data Pipeline

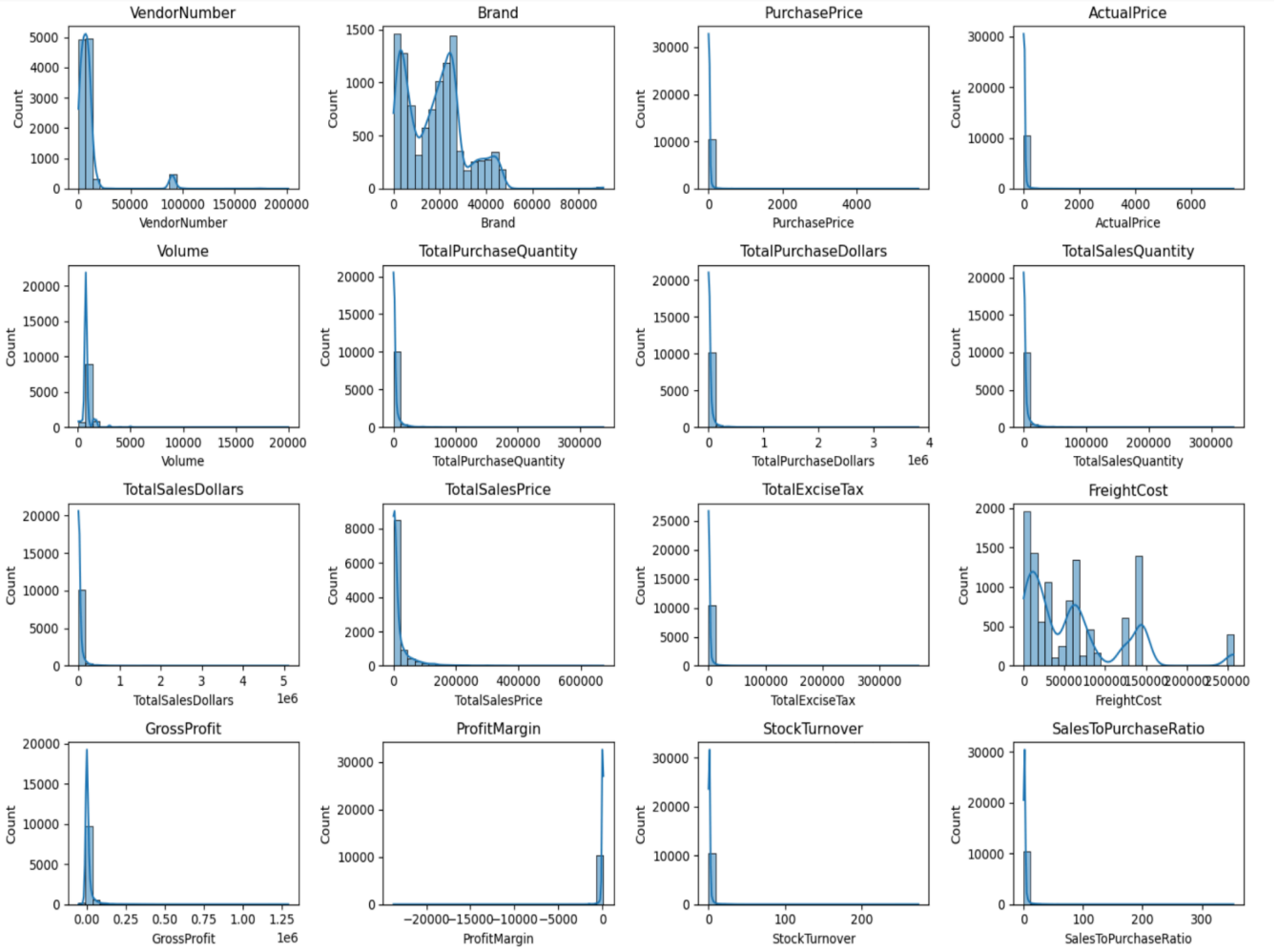
The project begins by loading raw CSV files into a SQLite database using Python. The data from multiple tables is integrated using SQL queries to create a consolidated Vendor Sales Summary table. After cleaning the data and engineering business metrics such as Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio, Exploratory Data Analysis (EDA) and statistical analysis are performed to uncover meaningful insights. Finally, the processed data is visualized in Power BI through an interactive dashboard that supports business decision-making and strategic recommendations.

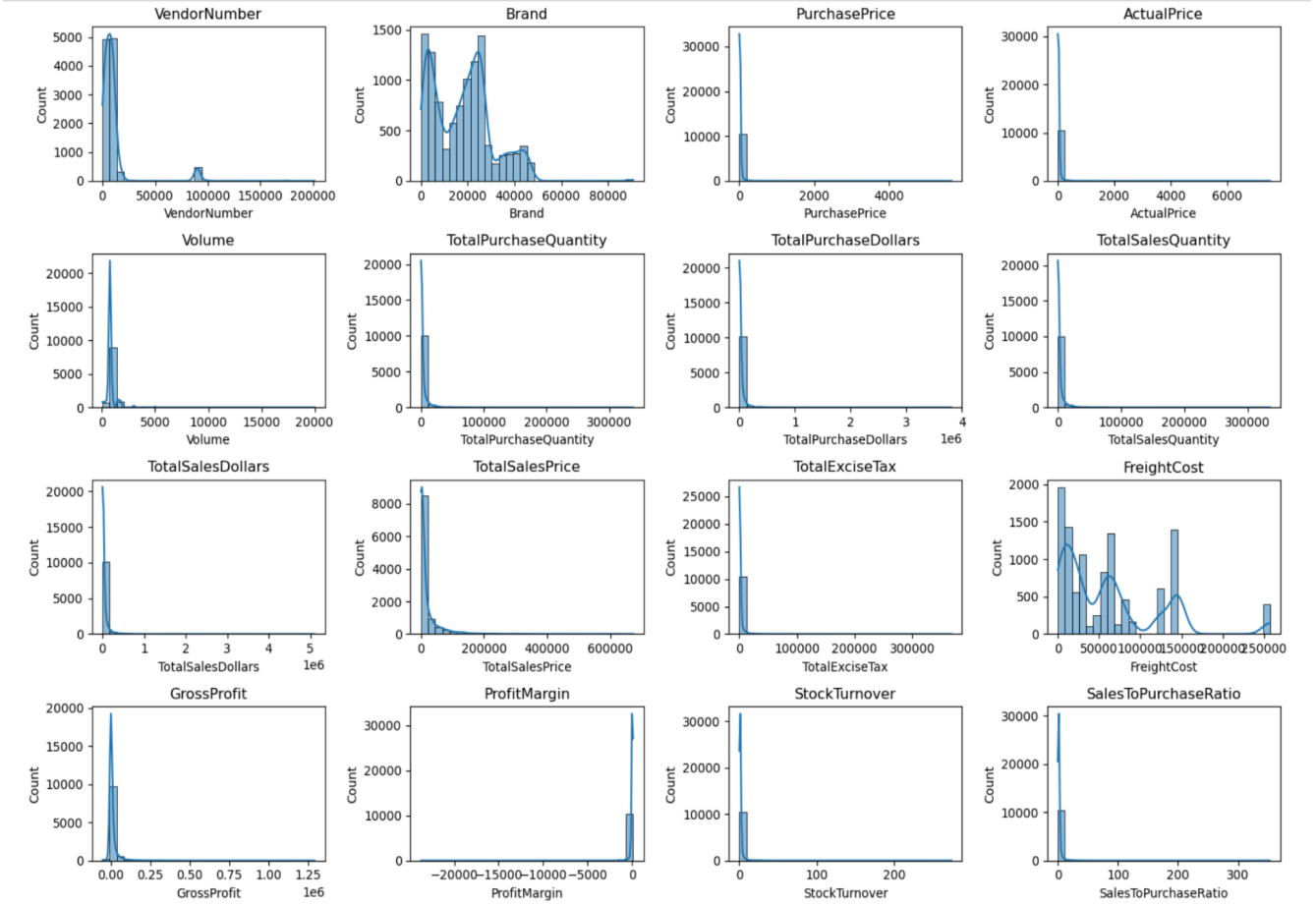


Exploratory Data Analysis Insights

Summary Statistics :

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02





Negative and Zero Values :

- **Gross Profit :** Minimum of -52,002.78, indicates potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.
- **Profit Margin :** Has a Minimum of -∞, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.
- **Total sales Quantity & Sales Dollars :** Some products show zero sales, indicating they are purchased but never sold. These may be slow moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard deviations :

- **Purchase & Actual Prices :** The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.
- **Freight Cost :** Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

- **Stock Turnover :** Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

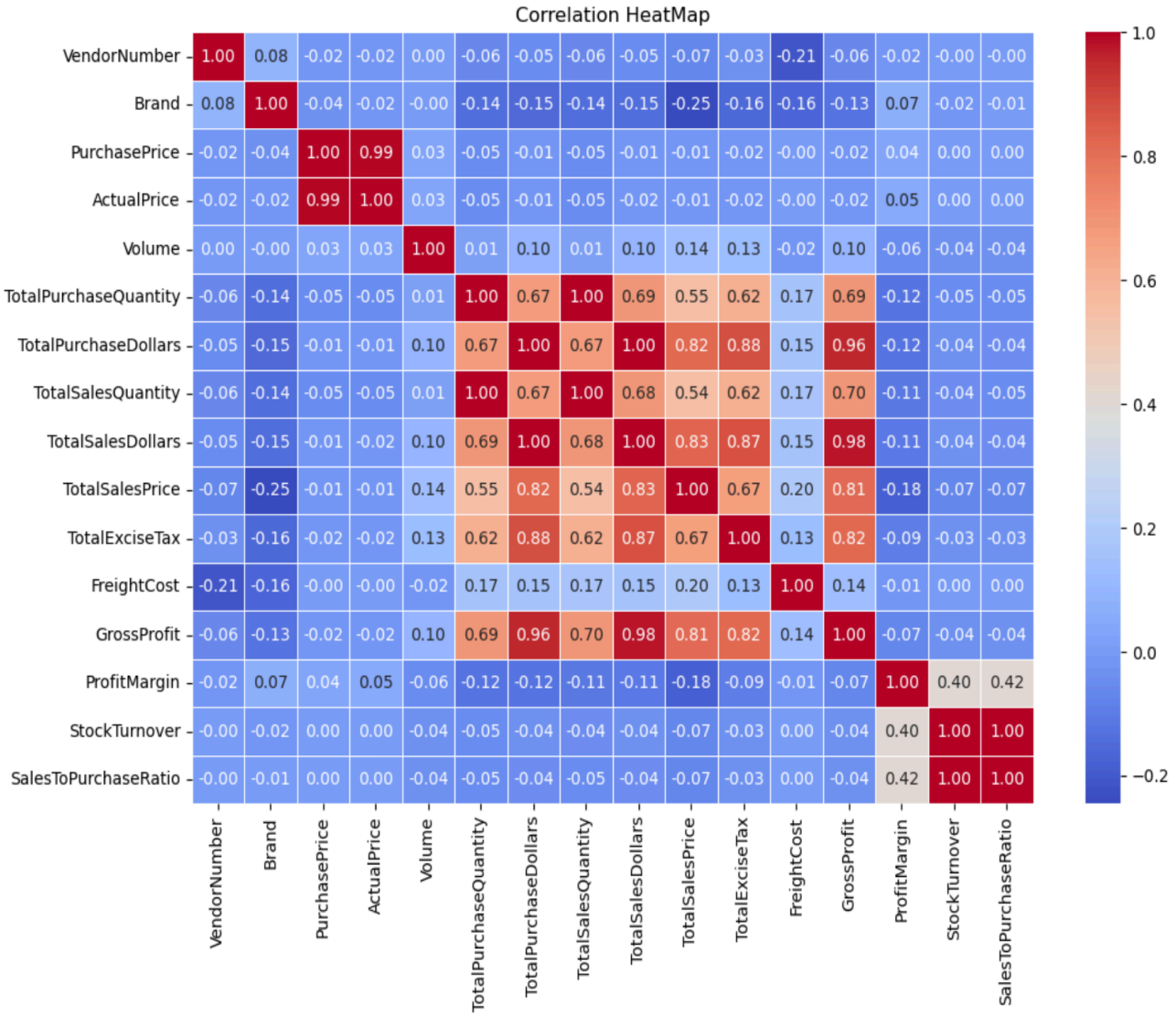
To enhance the reliability of the insights, we removed inconsistent data points where :

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (To ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

```
df = pd.read_sql_query(
    """
    SELECT * FROM vendor_sales_summary
    WHERE GrossProfit > 0
    AND ProfitMargin > 0
    AND TotalSalesQuantity > 0
    """,
    conn
)
```

Correlation Insights

- **Purchase Price vs Total Sales Dollars & Gross Profit :** Weak Correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.
- **Total Purchase Quantity vs. Total Sales Quantity :** Strong Correlation (0.999), confirming efficient inventory turnover.
- **Profit Margin vs. Total Sales Price :** Negative Correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.
- **Stock Turnover vs. Gross Profit and Profit Margin :** Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.



Research Objectives and Key Findings

1. Brands for Promotional or Pricing Adjustments :

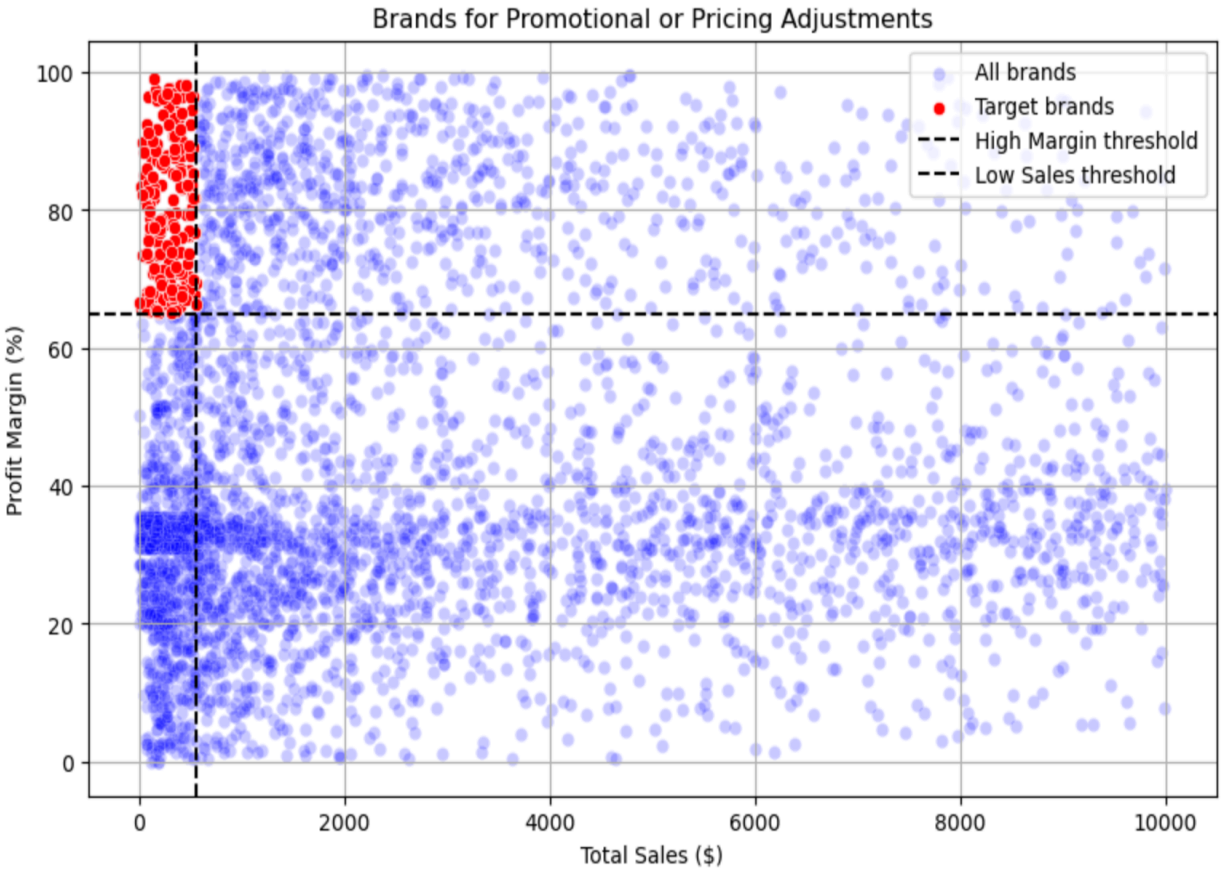
```
target_brands = brand_performance[
    (brand_performance['TotalSalesDollars'] <= low_sales_threshold) &
    (brand_performance['ProfitMargin'] >= high_margin_threshold)
]
```

198 Brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

```
--------------------------------------------------
Brands with Low Sales but High Profit Margins :
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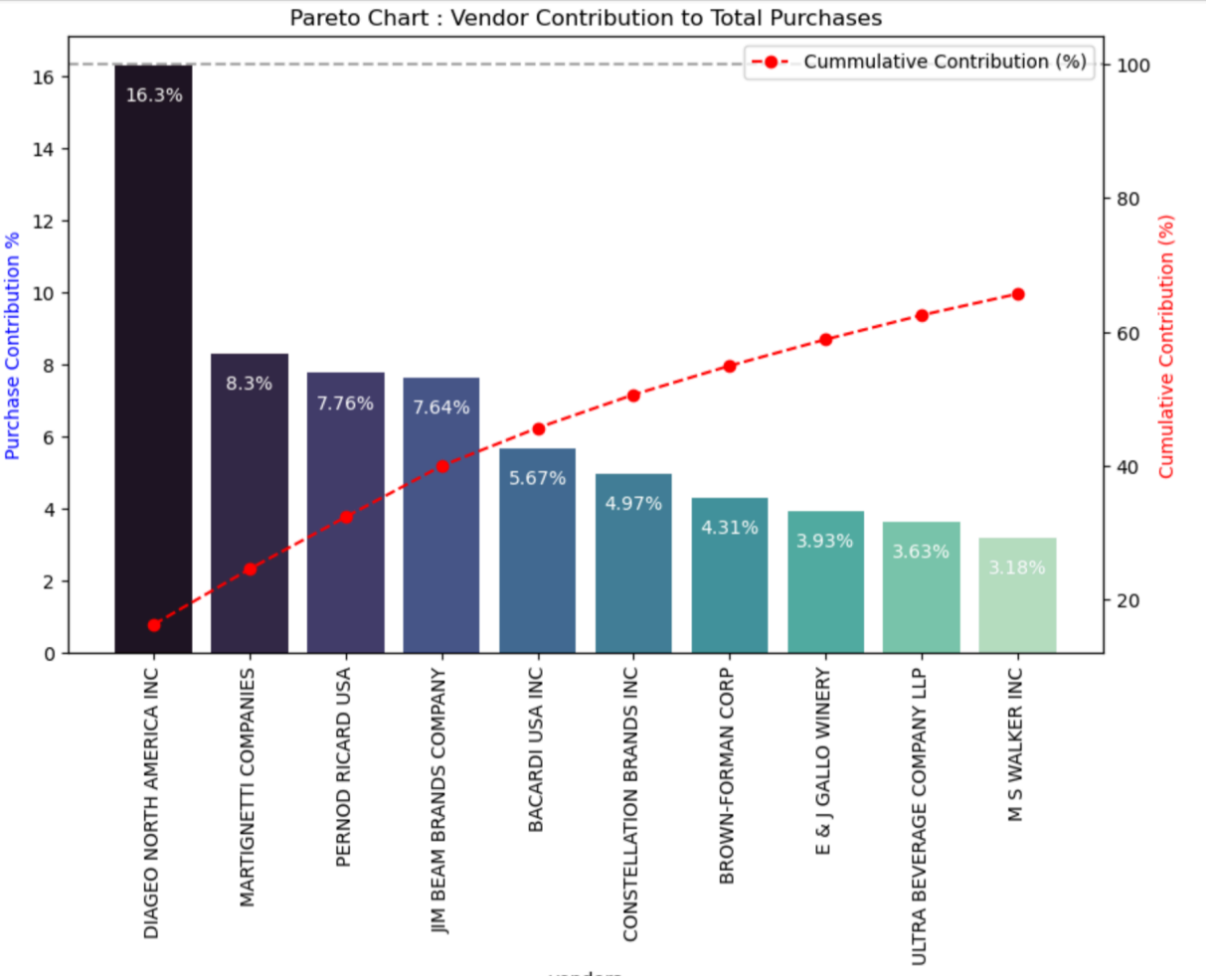
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

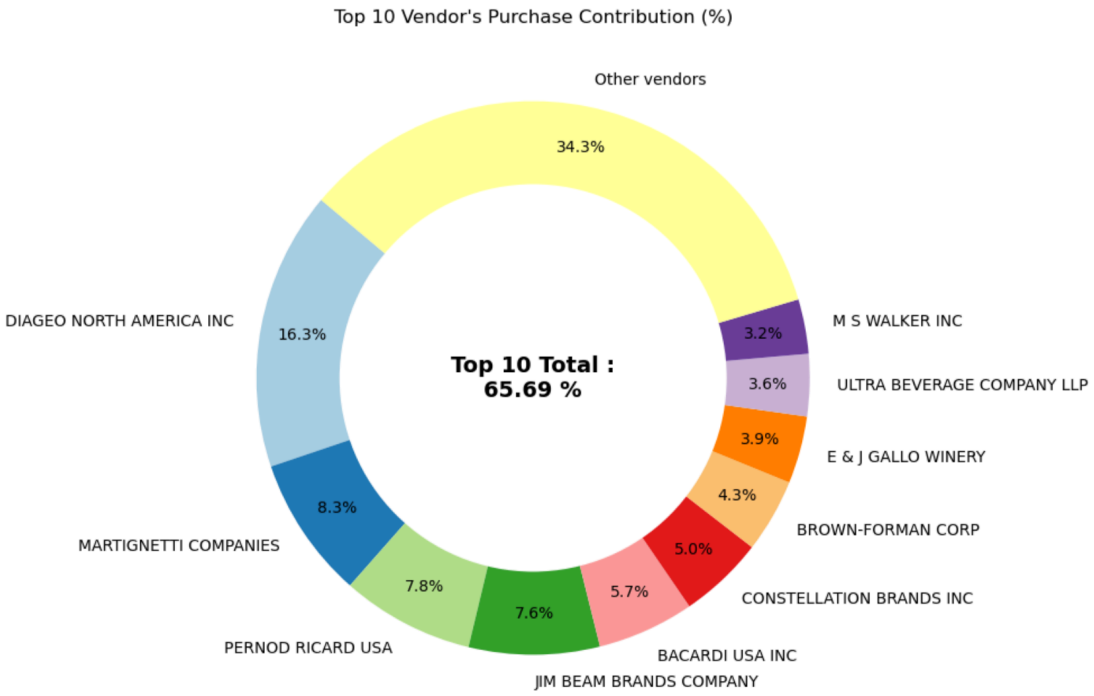
198 rows × 3 columns



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total Purchases, while the remaining vendors contribute only 34,31%. This over reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



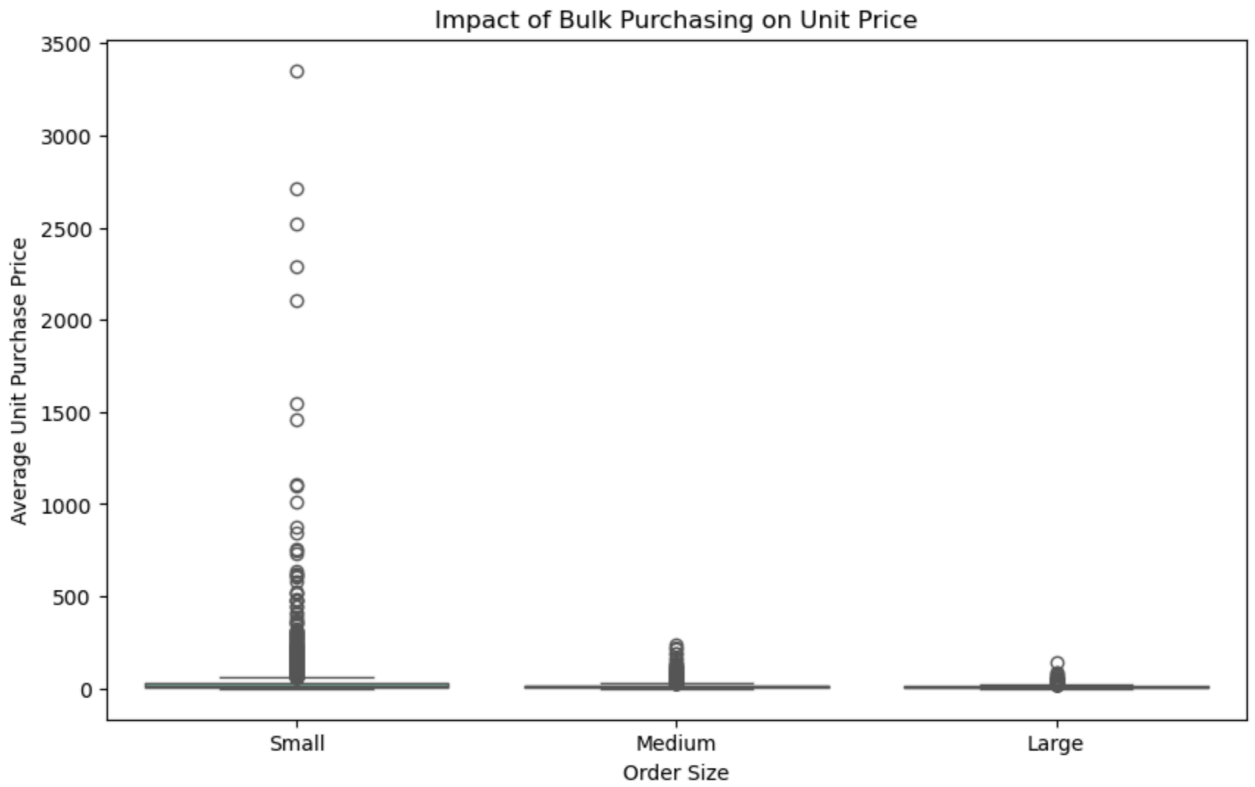


3. Impact of Bulk Purchasing on Cost Savings

- vendors buying in bulk (Large Order Size) get the lowest unit price ($10.78 per unit), meaning higher margins if they can manage inventory efficiently.
- The price difference between Small and Large order is substantial (~ 72% reduction in unit cost)

	OrderSize	UnitPurchasePrice
0	Small	39.068186
1	Medium	15.486414
2	Large	10.777625

- This suggests that bulk pricing strategies successfully encourage vendors to purchase in larger volumes, leading to higher overall sales despite lower per unit revenue.



4. Identifying Vendors with Low Inventory Turnover

Total Unsold Capital : $2.71M

- Slow moving inventory increases storage costs, reduces cash flow efficiency, and affects overall Profitability.

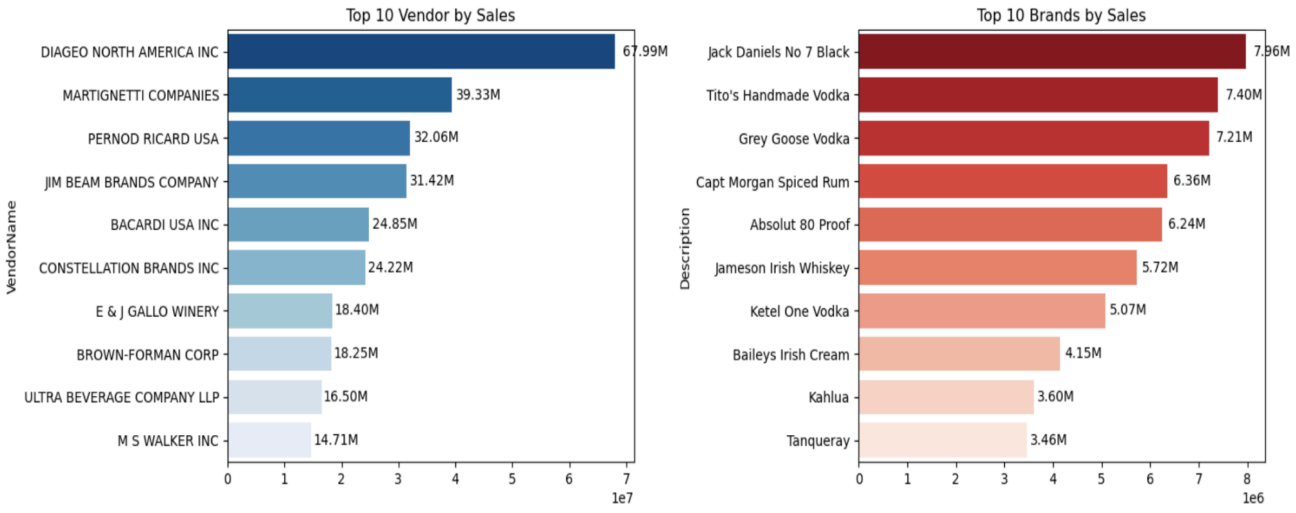
- Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

```
df["UnsoldInventoryValue"] = (df["TotalPurchaseQuantity"] - df["TotalSalesQuantity"]) * df["PurchasePrice"]
print('Total Unsold Capital : ', format_dollars(df["UnsoldInventoryValue"].sum()))
```

```
Total Unsold Capital :  2.71M
```

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

5. Vendors and Brands demonstrate the Highest Sales Performance



6. Profit Margin Comparison : High vs. Low Performing Vendors

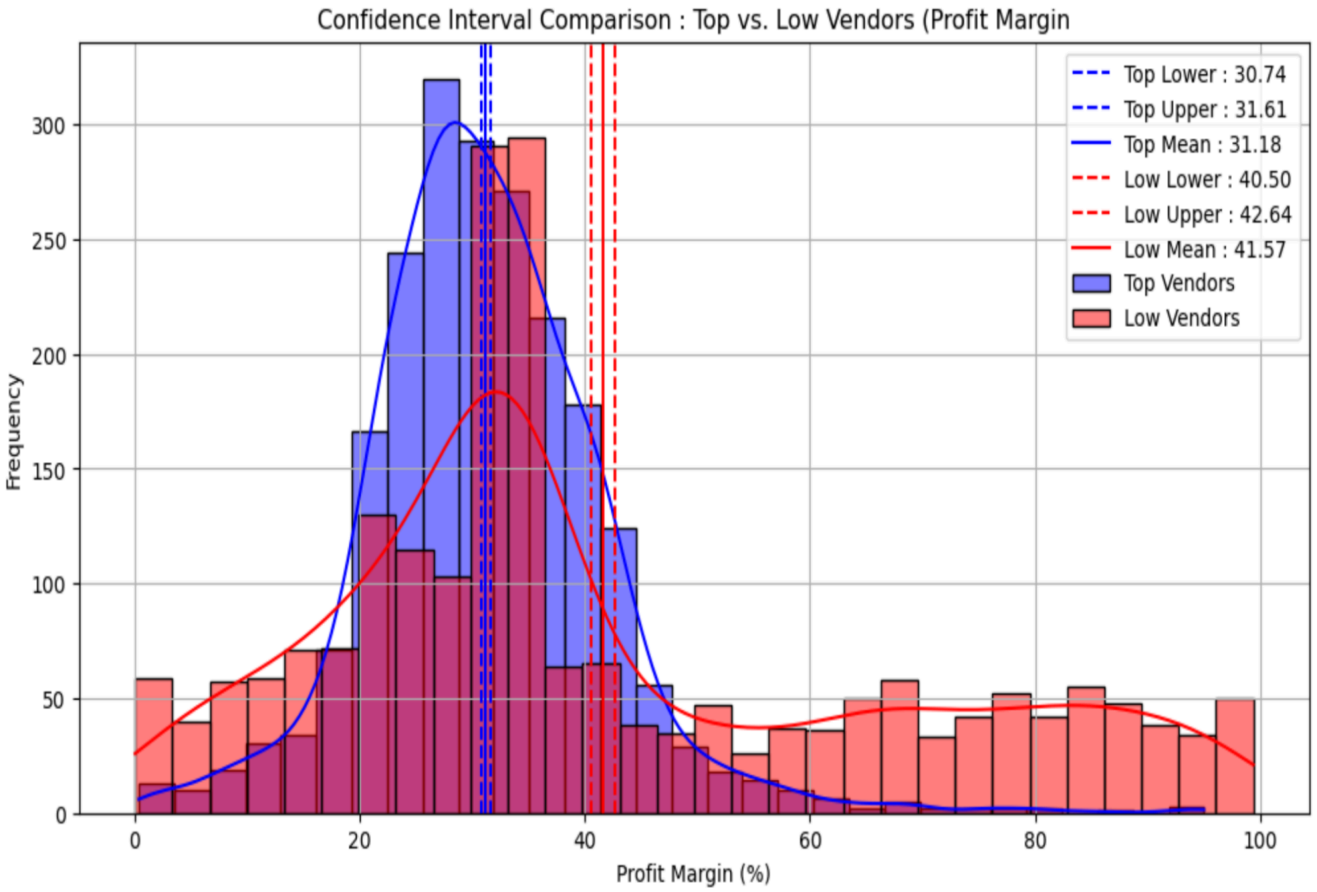
Top Vendors 95% CI : (30.74, 31.61), Mean : 31.18

Low Vendors 95% CI : (40.50, 42.64), Mean : 41.57

Low performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insight :

- The confidence interval for low performing vendors (40.50% to 42.64%) is significantly higher than that of top performing vendors (30.74% to 31.61%).
- This suggests that vendors for low sales tend to maintain higher profit margins, potentially due to premium pricing or lower operational costs.
- **For High Performing Vendors :** If they aim to improve profitability, they could explore selective price adjustments, cost optimization, or building strategies.
- **For Low Performing Vendors :** Despite higher margins, their low sales volume might indicate a need for better marketing, competitive pricing, or improved distribution strategies.



7. Statistical Validation of Profit Margin Differences

Hypothesis Testing :

- H_0 (Null Hypothesis) : There is no Significant difference in the mean profit margins of top performing and low performing vendors.
- H_1 (Alternative Hypothesis) : The mean profit margins of top performing and low performing vendors are significantly different.

Result :

- T-Statistics : -17.6695, P-Value : 0.0000
- Reject H_0 : There is a significant difference in profit margins between top and low performing vendors

Implication : High margin vendors may benefit from better pricing strategies, while top selling vendors could focus on cost efficiency.

KPI Formula

KPI	Formula	Business Purpose
Total Sales ($)	Σ Sales Dollars	Represents the total revenue generated from product sales.
Total Purchase ($)	Σ Purchase Dollars	Represents the total amount spent on purchasing inventory from vendors.
Gross Profit ($)	Total Sales Dollars − Total Purchase Dollars	Measures the profit earned after deducting purchase costs from sales revenue.
Profit Margin (%)	(Gross Profit / Total Sales Dollars) × 100	Indicates the percentage of revenue retained as profit. A higher value represents better profitability.
Stock Turnover	Total Sales Quantity / Total Purchase Quantity	Measures how efficiently inventory is sold. Higher values indicate faster inventory movement.
Sales-to-Purchase Ratio	Total Sales Dollars / Total Purchase Dollars	Compares sales revenue against procurement costs to evaluate purchasing efficiency.
Unsold Capital ($)	Purchase Value − Sales Value of Unsold Inventory	Represents the capital tied up in unsold inventory, helping identify slow-moving stock.

Final Recommendations

- Re evaluate pricing for low sales, high margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimise slow moving inventory by adjusting purchases quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

Conclusion

This project successfully analyzed vendor performance by integrating multiple business datasets and applying data cleaning, feature engineering, exploratory data analysis, and statistical techniques. The analysis provided valuable insights into vendor profitability, purchasing behavior, inventory efficiency, and sales performance.

The results highlighted several key business challenges, including dependence on a small number of vendors, slow-moving inventory, pricing inefficiencies among low-performing brands, and opportunities to leverage bulk purchasing for cost savings. Statistical analysis confirmed a significant difference in the profit margins of high-performing and low-performing vendors, supporting data-driven decision-making.

Based on these findings, the project recommends optimizing pricing strategies, reducing vendor dependency, improving inventory turnover, and enhancing procurement planning to increase overall profitability.

Overall, this analysis demonstrates how Python, SQL, and statistical methods can transform raw business data into actionable insights, enabling organizations to make informed strategic decisions and improve operational performance.